University of Kufa
Faculty of Computer Science and Mathematics
Department : Computer Science
Course Name : Software Engineering

Academic Year : 2021 / 2022
Course Stage / Level : 1st
Date of Exam : 15 / 5 / 2022
Time : 1 Hours

Note: Answer all questions.

Q1: Choose between brackets (5 marks)

1. If you create an array A= new int [3] you should think of it as a "*Array*" with----------

(a.3 rows and 1 column b. 1 row 3 columns c. 3 rows 3 columns d. 1 row 1 column)

2. To removes any spaces at the beginning or end of a string using --------- method.

a.trim() b. equals() c. endsWith() d.startsWith())

3. charAt(i) Returns the character at that position (i) in the string as -------.

(a. int b.string c. char d. boolean)

4. The action of changing a data type into a different type is called-------.

(a. type Conversion b. break c. continue d. type casting)

5. The output of the following statement System.out.println(Math.ceil(50.7)); is----------.

(a. 50 b. 51 c. 51.0 d. 50.0)

Q2 : Write down a Java program to read a 2-dimensional array x[4][4] then perform each of the following:(15 marks)

1. Find the maximum and minimum in the third row.
2. print positive numbers only in the main diagonal. فقط رئيسي موجب
3. Replace the upper triangle elements with 1.

-1	3	5	6
5	-2	4	6
3	4	5	7
1	7	9	-3

Q3. Find the output of the following code: (10 marks)

A)

```
for(i=1; i<=5; i++)
  { for(j=1; j<=i; j++)
      System.out.print(i+j);
      System.out.println();
  }
```

1 2 3 4 5
2
3 4
4 5 6
5 6 7 8
6 7 8 9 10

B)

```
int[] Array = {2, 5, 1, 9, -11};
System.out.println(Array.length);
System.out.println(Array[-2]);
```

5

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University of Kufa
Faculty of Computer Science and Mathematics
Department : Computer Science
Course Name: Advanced programming language

Academic Year : 2021 / 2022
Course Stage / Level : 1st
Date of Exam : 15 / 5 / 2022
Time : 1 Hours

Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextInt() method, which is used to read string.
2. floor(x) method Rounds x to the smallest integer not less than x?
3. To access the elements of an array we use the { } operator.
4. equals method Removes any spaces at the beginning or end of a string.
5. The output of charAt(i) method is int.

Q2 Fill in the blanks (5 marks)

1. The ---------- usually used with the switch-case statement; however, it can be used with a loop statement where it terminates the loop.
2. the string index begins at ----------
3. length Returns the length of the string as ----------
4. int a = (int)18.7; a=----------
5. A ---------- is an indexed list of values that all have the same type.

Q3 What is the output when each of the following statement is executed?(5 marks)

str = " Computer Science";

a) System.out.println(str.trime());

b) System.out.println(str.startsWith ("Java"));

c) System.out.println(Math.abs(7.5));

d) System.out.println(Math.floor(7.5));

e) System.out.println (Math.min(5,6));

Q4: Write a Java program to search an element in an array(10 marks).

Q5: Write down a Java program to reverse the order of elements in an array, for example: A = {12, 0, 3, -4, 7} becomes A= {7, -4, 3, 0, 12}.(5 marks).

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Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextLine() method, which is used to read int.
2. ceil(x) method Rounds x to the largest integer not greater than x.
1. To access the elements of an array we use the [] operator.
2. toUpperCase method Returns a copy of the original string, all lower case.
3. The output of Substring(i,j) method is int.

Q2 Fill in the blanks (5 marks)

1. In Java, the class --------- provides pre-written mathematical functions to perform various math calculations.
2. The method ---------- used to find the natural logarithm of x (base e).
3. System.out.println(Math.ceil(45.87)); output is----------
4. The method used to compare two strings without considering the case of small or capital letters is ----------
5. The ---------- statement skip the current iteration of a loop and proceeds directly to the next iteration.

Q3 What is the output when each of the following statement is executed?(5 marks)

str = " university of kufa ";

a) System.out.println(str.length());
b) System.out.println(str.startsWith ("Java"));
c) System.out.println(str.Substring(1,4));
d) System.out.println(Math.ceil(7.5));
e) System.out.println (Math.abs(5));

Q4: Write a Java program to sort the items of an array in ascending order(10 marks).

Q5: Write a Java program that performs the following operations on an one-denominational array of type double: Array size [10] (5 marks)

a) Print out odd numbers in the array.

b) Display the values of the third column.

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University of Kufa
Faculty of Computer Science & Mathematics
Department: Computer Science
Course Name: Advanced Programming

Academic Year 2020-2021
Course Stage/Level: 1st
Date of Exam: 10/06/2021
Time: One hour

Notes: Attempt all questions

Q1> Answer each of the following statements: (7 marks)

1) What is the output when each of the following statement is executed?

```
String str = "    KUFA UNIVERSITY   ";
System.out.println(str.trime());
System.out.println(startsWith("kufa"));
```

2) The method used to compare two strings without considering the case of small or capital letters is .....

3) The ..... statement skips the current iteration of a loop and proceeds directly to the next iteration.

4) Cast the following float value into integer: int x = 120.5f; .....

5) Correct the error of the following set of code:

```
int count=-11.5;
System.out.println(Floor(Count));
System.out.println(abs(Count));
```

Q2> Write a Java program to find the square root of the smallest number out of five double numbers. (4 marks)

Q3> Write a Java program to read two names and print them in an alphabetical order. (4 marks)

Good Luck

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Academic Year : 2021 / 2022
Course Stage / Level : 1st
Time : 2 Hours
Final exam: First attempt
First sample

Note: Answer all questions.

Q1 TRUE OR FALSE(Answer 10 only) (10 marks)

1. If you create an array int A= new int [3][4], you should think of it as a "matrix" with 4 rows and 3 columns.
2. abs(x) method return the absolute value of x.
3. In array, the index must start by 1.
4. To access the elements of an array we use the { } operator.
5. The output of Substring(i,j) method is string.
6. Method contains a set of instructions that perform well defined task and only runs when it called.
7. sqrt(),is a type of user-defined method.
8. trim method Removes any spaces at the beginning or end of a string.
9. Arrays that require two indexes to identify a particular element called one dimensional array.
10. Implicit Type Conversion used when convert a larger type to the smaller type size.
11. toUpperCase method Returns a copy of the original string, all lower case.

Q2: What is the output when each of the following statement is executed? Choose two only(4 marks)

A.
```
String  start = "Talk is cheap. ";
String middle = "Show me the code. ";
String end = "- Linus Torvalds";
String result = start + middle + end;
   System.out.println(result);
   System.out.println(result.charAt(7));
```
.....

B.
```
double num1= 45.78;
System.out.println(Math.ceil(num1));
    System.out.println(Math.floor(num1));
```
.....

C.
```
double[] Array = {-2.0, 5.6 , -1.3, -9.0, 11.8};
System.out.println(Array.length);
System.out.println(Array[1]+Array[3]);
System.out.println(Array[2]);
```

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Q3: Fill in the blanks for 6 only(6 marks)

1. The number used to refer to a particular array element is called the element's……2……
2. If you create an array int A= new int [10] you should think of it as a "*Array*" with……. rows and ……. Columns.
3. The mathematical function used to find the largest number is ……
4. The method ---------- used to find the square root of x.
5. …….. method accepts a string and add it to the end of the original string.
6. The action of changing a data type into a different type is called ……
7. Cast the following float value into integer: int x = 120.5f; ……
8. nextFloat() is a type of _____ methods.

Q4: With a 2-D array ~~of ten integer~~ numbers, write a Java program to find the following: **choose three only** (12 marks)

a) The average of even numbers.
b) Print prime numbers only.
c) Print negative numbers only in the Secondary diagonal.
d) Replace the main diagonal elements by 1

Q5: Write a Java program has methods that perform the following: **choose two only** (8 marks)

a) Return the absolute value of a number.
b) Test whether a number is even and returning true if it is.
c) Takes a string as input and print it out.

Good Luck

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